

Original Article

Developmental Toxic Effects of Antifungal Flusilazole Administered by Gavage to Mice

Amina T. Farag* and Hala H. Ibrahim

Department of Pesticide Chemistry and Toxicology, Faculty of Agriculture, Alexandria University, Egypt

BACKGROUND: The developmental toxicity of flusilazole was studied in CD-1 mice after oral administration. **METHODS:** Pregnant mice were given flusilazole at doses of 0 (corn oil), 10, 20, and 40 mg/kg/day, by gavage, on gestational days (GD) 6–15. **RESULTS:** Maternal toxicity, as evidenced by reduction in body weight gain and signs of toxicity, was observed at the middle- and high-dose groups. No significant incidence of resorptions or death was observed in any of dose groups. There was a pronounced reduction in fetal weight, which was significantly lower than control from 20 and 40 mg/kg/day. There was no significant increase in the incidence of fetuses with external or visceral malformations in any of dose groups, but there was a significant increase in the incidence of skeletal malformations was observed at 20 and 40 mg/kg/day. **CONCLUSIONS:** The results of this study reported marked maternal toxicity, growth retardation, and skeletal abnormalities in the mid- and high-dose groups. It seems likely that marked maternal toxicity contributed to the observed alterations in fetal growth retardation and skeletal development. The no-observed-effect level in the present study for maternal and developmental toxicity was 10 mg/kg/day. *Birth Defects Res (Part B)* 80:12–17, 2007. © 2006 Wiley-Liss, Inc.

Key words: *developmental toxicity; flusilazole; mice; skeletal malformations; teratogenic effects*

INTRODUCTION

Triazole derivatives are antimycotic compounds used widely in human and veterinary therapy and agriculture. Interference with steroid biosynthesis and with formation of fungal cell walls has been identified as the mechanism of these compounds (Vanden et al., 1984). Because of their action, these compounds are used successfully as systemic agriculture fungicides (against mildews and rusts of cereal grains, fruits, vegetables, and ornamentals) and against systemic and topical fungal diseases in humans and domestic animals (Menegola et al., 2001).

Flusilazole, a synthetic azole antifungal agent, is an economically important agricultural chemical that is used widely on crops such as wheat, barely, and orchard fruits (Buchenauer, 1987). Although flusilazole has low toxic effects (the reported acute oral LD₅₀-value for the female mice is 1000 mg/kg/day) (Wylie et al., 1985), it has been reported as a teratogen in experimental animals (Food and Agriculture Organization, 1989; Vergieva, 1990). Animal models in vitro and in vivo have been used to investigate the teratogenicity of flusilazole. Specific teratogenic effects on the branchial arch apparatus has been described in culture whole rat embryos (Menegola et al., 2001). Flusilazole has been tested for in vivo teratogenic effects in rats and rabbits (Lamontia et al., 1984; Solomon et al., 1984; Food and Agriculture Organization, 1989; Vergieva, 1990; Hollingsworth, 1995).

When pregnant rats were treated with 250 mg/kg/day during gestation days (GD) 7–16, an increased occurrence of fetal skeletal variation, including misaligned sternebra, extra ossification centers in ribs, and ossification delay in sternebra, were noted (Lamontia et al., 1984). Cleft palate, absence of renal papilla, large renal pelvis, and small renal papilla has been also reported in the fetuses after exposure to 250 mg/kg/day during GD 7–16 (Lamontia et al., 1984). Doses ranging from 1/5 to 1/50 mg/kg/day of LD₅₀-value equal to 1272 mg/kg/day resulted in increased fetal abnormalities exophthalmus, hypognathia, microglossia, and cleft palate in female rats treated on GD of 9, 10, 11, or 13 (Vergieva, 1990). Furthermore, oral doses of flusilazole given to pregnant rats produced congenital defects such as protruding eyes, twins of unequal size joined at the jaw, and grossly enlarged tongues (Hollingsworth, 1995). A common severe teratogenic intrinsic property for flusilazole has been found, acting through alteration of the normal hindbrain developmental pattern (Menegola et al., 2005). No evidence of teratogenicity was found after exposure of pregnant rabbits to 1.9, 4.8, and 10.1 mg

*Correspondence to: Amina Tolba Farag, PhD, Department of Pesticide Chemistry and Toxicology, Faculty of Agriculture, (El-Shatby), University of Alexandria, Egypt. E-mail: aminafarag2002@yahoo.com
Received 22 July 2006; Accepted 20 September 2006

Published online in Wiley InterScience (www.interscience.wiley.com)
DOI: 10.1002/bdrb.20098

flusilazole/kg/day during Days 7–19 after insemination (Solomon et al., 1984). Although flusilazole remains one of the most widely used fungicides in the world, there are limited current investigations of its teratogenic capacity, especially at low dose levels in female mice. In the present study, the possible developmental toxicity of flusilazole was investigated at the dose levels lower than those reported to produce developmental toxicity in previous published studies (Food and Agriculture Organization, 1989; Vergieva, 1990). These doses are equal to 1/100, 1/50, and 1/25 of LD₅₀-value (Wylie et al., 1985).

MATERIAL AND METHODS

Chemical

Technical grade flusilazole (99% purity) was obtained from Center Laboratory of Pesticides (El-Doky, Cairo, Egypt).

Animals and Conditions

Sixty male and 120 female ICR (CD-1) mice, approximately 10 weeks old, were obtained from Faculty of Science, Department of Zoology, Alexandria University, Alexandria, Egypt. All mice were examined for health status and acclimated to the laboratory environment for 2 weeks before use. Temperature was maintained at $23 \pm 2^\circ\text{C}$, and relative humidity at approximately 50%, with a 12 hr:12 hr light:dark photoperiod. Animals were housed in stainless steel cages and given standard diet and water ad lib throughout the study. Adult virgin female mice were mated with adult males (one male/two females). The day that a vaginal plug was found was considered Day 0 of gestation. Bred females were assigned randomly to treatment groups according to an approximately equal mean body weight. Groups of 30 bred mice each were given flusilazole by oral gavage on GD 6–15 at the dose levels of 0, 10, 20, and 40 mg/kg/d flusilazole in corn oil as the vehicle. The administered volume of each dose was 0.4 ml/kg body weight per day, adjusted daily for recorded body weight changes during the treatment period.

Maternal Observations

Females were examined daily throughout the experimental period and signs of toxicity were recorded. Maternal body weight was recorded on GD 0, 6, 9, 15, and 18. Feed consumption was recorded on GD 6, 9, 12, 15, and 17. Weights of maternal liver, kidneys, brain, spleen, heart, and placenta were recorded at the time of Cesarean section on GD 18.

Fetal Observations

Mated pregnant females were anaesthetized using diethyl ether on GD 18. The uterine horns were examined for the number and location of fetuses and resorption sites. Fetuses were removed, weighed, sexed, and evaluated for external abnormalities. The uteri of apparently non-pregnant mice were stained with 10% sodium sulfide (Salewski, 1964) and examined for evidence of implantation sites. Fifty percent of the fetuses were fixed in Bouin's solution for razor blade sectioning (Wilson and Warkany, 1965). The remaining

fetuses were fixed in alcohol, double stained with alizarin red S for ossified bone and Alcian blue for cartilage, and cleared in 2% KOH and glycerin (Peters, 1965; Hayes, 1994).

Statistical Evaluation

Maternal toxicity was estimated from maternal body weights, weight gain, feed consumption, and absolute and relative organ weights. One-way analysis of variance procedure (ANOVA) followed by Tukey's multiple comparisons with the litter as the treatment unit was used to compare malformation rates between the treated groups. The maternal body weight on GD 6 was used as a covariant (ANCOVA) for comparing weights of the treated groups with those of the control. The proportions of resorbed litters were compared using Z-test for differences of proportions (Walpole and Meyers, 1978) with a modified Bonferroni correction (Simes, 1986). Fetal weights were analyzed using a nested ANOVA followed by Tukey's multiple comparisons (Winer, 1971). Percentages of resorbed fetuses were evaluated with pairwise Mann-Whitney tests with a modified Bonferroni correction to compare each treatment to the control (Lehmann, 1975). Sex ratio and litter size data were compared with the Kruskal-Wallis test (Lehmann, 1975; Norusis, 1994). The statistical analysis were conducted using a nominal $\alpha = 0.05$.

RESULTS

Maternal Observations

Maternal clinical signs of toxicity. There were no deaths or abortions during the course of the present study. Signs of toxicity in the form of convulsions and hyperactivity were noted in dams at 20 and 40 mg/kg/day flusilazole compared with the control and other treated groups. These signs appeared on GD 9 (Day 4 of treatment) in 60–80% of treated dams with the middle and high dose groups, respectively. These signs progressed in the same animals throughout the second half period of the treatment. No effects were noted in the general appearance of animals in the control and 10 mg/kg/day treated group.

Maternal body weight and weight gain. Maternal body weight gains are presented in Figure 1. There were statistically significant reductions in the maternal body weights at GD 9–18 for the 20 and 40 mg/kg/day dose groups compared with the control. Maternal weight gain was also reduced in the same dose groups among the entire period of gestation compared with the control group. There were no recorded changes in maternal weight gains in the low dose group compared with the control group. No appreciable changes in feed consumption were noted among any of the dose groups compared with the control group (data not presented).

Maternal organ weights. The absolute weights of liver, kidneys, brain, spleen, heart, and placenta were significantly decreased in the 20 and 40 mg/kg/day dose groups compared with control and other treated groups. No significant changes were noted in the relative organ weights among any of the treated groups (data not shown).

Fetal Observations

Developmental parameters and fetal weights are summarized in Table 1. The number of the implants per litter was not altered significantly in the flusilazole treated groups. No significant changes in the number of resorptions live and dead fetuses in any of the treated groups compared with control group. Although no significant difference was observed in the sex ratio of

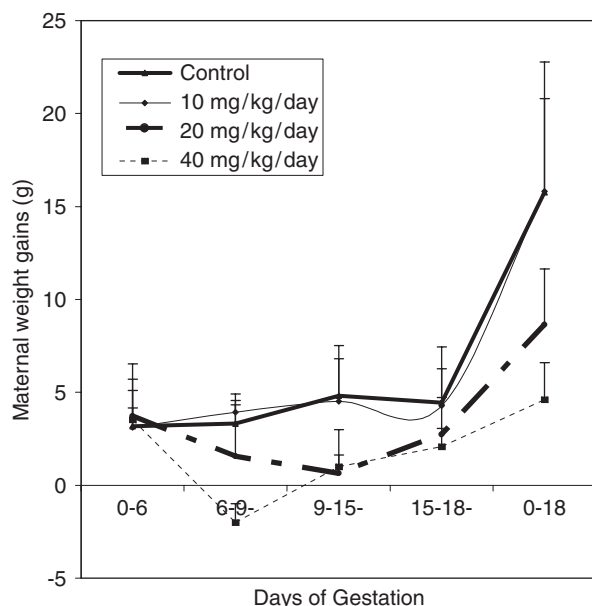


Fig. 1. Maternal weight gain (g) in mice after exposure to flusilazole on GD 6–15. Data are presented as mean $\pm$ SD. * $P \leq 0.05$; $P \leq 0.01$.

the fetuses in any of the treated groups, fetal weights for mice fetuses were significantly lower than those in the control group for the 20 and 40 mg/kg/day treated groups.

Table 2 depicts the number of malformations and variations in the surviving fetuses from dams exposed to flusilazole. No significant external or visceral malformations were observed in any of the treated groups compared with the control group. Several external malformations were observed scattered throughout the dose levels. Exophthalmos (unilateral) and absence of eyelids (bilateral) was observed in one fetus in the 20 mg/kg/day group and in two fetuses in the 40 mg/kg/day dose groups. One case of cleft palate (soft-palate) was observed in 40 mg/kg/day dose group compared with a single case in the control. Regarding to the visceral malformations, large renal pelvi was observed in one and two cases in the middle and high treated groups compared with a single case in the 10 mg/kg/day and control groups. Visceral variations such as subcutaneous hemorrhage was observed in one, one, two, and three fetuses in the control, 10, 20, and 40 mg/kg/day groups, respectively. Bladder foci were only observed in one fetus in the high dose group. Skeletal malformations and variations were observed across the high and middle dose groups compared with the 10 mg/kg/day and control groups. Skeletal evaluation showed large number of malformations and variations, primarily involving the axial skeleton. They included skeletal malformations: absence of sacrum, fused, wavy, and absent of ribs, shortening of long bones, absence of metatarsals and metacarpals, and absence of phalanges; and variations: poorly ossified interparietal, poorly ossified thoracic centrum, cranial retardation, and retardation of pelvic bones. However, these incidences of skeletal malformations and variations were increased in the 20 and 40 mg/kg/day dose groups.

Table 1
Developmental Toxicity in Mice Fetuses After Exposure to Flusilazole on GD 6–15

	Flusilazole (mg/kg/day)			
	0	10	20	40
No. of females	30	30	30	30
No. of pregnant (%) ^a	28 (93)	27 (90)	26 (87)	25 (83)
No. of litters	28	27	26	25
No. of implantation/litter	10.5 $\pm$ 2	10.2 $\pm$ 4	10.7 $\pm$ 3	9.8 $\pm$ 2
% postimplantation loss ^b	2.0 $\pm$ 7	2.0 $\pm$ 5	3.0 $\pm$ 5	4.0 $\pm$ 8
Live fetuses/litter (%) ^c	10.3 $\pm$ 0.3 (98)	10.0 $\pm$ 0.2 (98)	10.4 $\pm$ 0.4 (97)	9.4 $\pm$ 0.9 (96)
Dead fetuses/litter (%) ^c	0.1 $\pm$ 4 (1)	0.1 $\pm$ 3 (1)	0.1 $\pm$ 4 (1)	0.1 $\pm$ 4 (1)
Early resorptions/litter (%) ^c	0.1 $\pm$ 8 (1)	0.1 $\pm$ 4 (1)	0.1 $\pm$ 3 (1)	0.2 $\pm$ 2 (2)
Late resorptions/litter (%) ^c	0.0 $\pm$ 0 (0)	0.0 $\pm$ 0 (0)	0.1 $\pm$ 2 (1)	0.1 $\pm$ 7 (1)
Litters with resorption (%)	2 (7)	2 (7)	2 (8)	3 (12)
Litters totally resorbed (%)	0	0	0	0
Sex ratio (M/F)/litter	1.1 $\pm$ 2	1.1 $\pm$ 2	1.1 $\pm$ 2	1.2 $\pm$ 1
Fetal body weight (g)/litter	3.33 $\pm$ 0.15	3.05 $\pm$ 0.11	1.98 $\pm$ 0.15*	1.22 $\pm$ 0.12*

Data are presented as $\pm$ mean SD

^aNumber of females detected as being pregnant by visual inspection of the uterus/total number of females with vaginal plugs $\times$ 100. Sodium sulfide staining was carried out on all uteri without gross evidence of pregnancy but did not disclose resorption sites in any of the evaluated dams.

^b[(no. of implants – no. live fetuses)/no. of implants] $\times$ 100.

^cPercent of implantations.

*Significantly different from control at $P < 0.01$.

Table 2
Incidence of Fetal Malformations and Variations in Mice After Exposure to Flusilazole on GD 6–15

	Flusilazole (mg/kg/day)			
	0	10	20	40
Total no. of fetuses (litters) examined ^a				
External	288 (28)	259 (27)	270 (26)	245 (25)
Visceral	140 (28)	125 (27)	135 (26)	117 (25)
Skeletal	148 (28)	134 (27)	135 (26)	128 (25)
External malformations ^b				
Exophthalmos (unilateral)	0	0	1 (1)	2 (1)
Absence of eyelids (bilateral)	0	0	1 (1)	2 (1)
Cleft plate (soft plate)	1 (1)	0	0	1 (1)
Total no. of fetuses (%) with external malformations	1 (0.4)	0	2 (1)	5 (2)
Total no. of litters (%) with external malformations	1 (4)	0	2 (8)	3 (12)
Mean % fetuses with external malformations/litter	1.19 ± 3	0 ± 0	2.5 ± 5	2.6 ± 4
Visceral malformations ^b				
Large renal pelvic	1 (1)	1 (1)	1 (1)	2 (1)
Visceral variations ^b				
Subcutaneous hemorrhage	1 (1)	1 (1)	2 (1)	3 (2)
Bladder foci	0	0	0	1 (1)
Total no. of fetuses (%) with visceral malformations and variations	2 (1)	2 (2)	3 (2)	4 (3)
Total no. of litters (%) with visceral malformations and variations	2 (7)	2 (7)	2 (8)	3 (12)
Mean % fetuses with visceral malformations and variations/litter	1.32 ± 5	1.0 ± 3	1.54 ± 6	2.12 ± 8
Skeletal malformations ^b				
Absence of sacrum	0	0	8 (6)	12 (7)
Ribs/fused/wavy/absence	0/0/0	0/0/0	5 (3)/3 (2)/2 (2)	5 (5)/6 (5)/5 (5)
Shortening of long bones	0	0	7 (5)	10 (5)
Absence of metatarsals/metacarpals	2 (1)/2 (2)	2 (1)/2 (1)	8 (3)/5 (2)	9 (7)/10 (8)
Absence of phalanges	3 (2)	5 (4)	9 (3)	16 (12)
Skeletal variations ^b	0	0	8 (6)	12 (7)
Poorly ossified interparietal	1 (1)	0	9 (5)	14 (6)
Poorly ossified thoracic centrum	0	0	6 (6)	11 (5)
Cranial retardation	0	0	8 (5)	13 (6)
Retardation of pelvic bones	0	0	3 (2)	9 (5)
Total no. of fetuses (%) with skeletal malformations and variations	8 (5)	8 (6)	54 (40)*	76 (59)*
Total no. of litters (%) with skeletal malformations and variations	3 (11)	5 (19)	20 (80)*	22 (88)*
Mean % fetuses with skeletal malformations and variations/litter	5.96 ± 4	8.93 ± 6	44.46 ± 8*	55.04 ± 11*

^aOnly live fetuses were examined.

^bFetus may be represented more than once in listing of malformations.

*Significantly different from control at $P < 0.01$.

DISCUSSION

To our knowledge, few published studies have been reported about the effects of flusilazole on the developmental toxicity during organogenesis period of pregnancy in experimental animals (Food and Agriculture Organization, 1989; Vergieva, 1990). This is the first study to evaluate the potential developmental toxicity of flusilazole on the female mice during the organogenesis period. In this study, the no-observed-adverse-effect-level (NOAEL) obtained for developmental effects was 10 mg/kg/day. This dose was incapable of inducing maternal or developmental toxicity compared with the control and other treated groups. Doses of 20 and 40 mg/kg/day, however, were maternally and developmentally toxic, with significant appearance of maternal clinical signs of toxicity and pronounced decreases in maternal weight gains. These results are consistent with the data indicated that flusilazole had systemic maternal toxicity at dose levels up to 50 mg/kg/day (Lamontia et al., 1984; Food and Agriculture Organization, 1989; Vergiena, 1990). Decreased maternal body weight in the absence

of changes in feed consumption may be related to maternal systemic toxicity.

Although flusilazole was unaffected the number of resorptions, live and dead fetuses, and sex ratio in any of the treated groups, pronounced decreases in fetus growth were observed in the mid- and high dose groups. At 20 and 40 mg/kg/day, maternal body weight gains were decreased significantly relative to the control group at all intervals measured. Thus, decreased maternal body weight gains were noted before the marked growth phase in the fetuses, which occurs primarily during the later stages of gestation. Therefore, the considerably reduced fetal weights may have been caused at least in part by maternal toxicity.

There was no significant difference in the incidence of any external and visceral malformations in any of the treated groups. Several abnormalities were observed scattered throughout the dose levels. These results contradicts the data reported that when administered several triazole-derivatives to pregnant rats and rabbits by gavage during the organogenesis period, they were able to produce external and visceral defects (Machemer,

1976; Food and Agriculture Organization, 1981, 1983, 1989; Vergieva, 1990; Machera, 1995). The common malformations for all these substances were cleft plate and cranio-facial malformations. These external malformations were observed in the present study, but not in significant manner as described by the previous studies. Therefore, it could be expected that these patterns of external and visceral malformations might be observed in the present study by increasing the dose levels of exposure. On the other hand, treatment with flusilazole resulted in a greater number of skeletal malformations and variations. These results are consistent with the data indicated that flusilazole was able to produce severe skeletal variants and abnormalities (Food and Agriculture Organization, 1989; Viergieva, 1990).

Fetal growth retardation could be related to such pronounced skeletal malformations or variations (Aliverti et al., 1979; Ariyuki et al., 1980, 1982; Chahoud and Paumgarten, 2005). Fetal skeletal effects could be related to developmental delays, particularly in light of the effects on maternal and fetal body weights. The skull, vertebrae (arches and bodies), forelimb, hindlimbs, and sternum are among the latest areas to undergo osteogenesis, and developmental delays are often seen in these areas. Shortening of long bones may be related to decreased fetal size.

In the present study, although flusilazole could induce growth retardation and skeletal malformations or variations in the mid- and high-dose levels, it seems unlikely to be related to the fungicide itself. Taking together in vivo and in vitro results, flusilazole induced only slight signs of developmental retardation when tested in vitro at very high concentration levels (5000 μ M) (Menegola et al., 2001, 2005). In addition, the results of this study reported marked maternal toxicity, growth retardation, and skeletal abnormalities in the mid- and high-dose groups. It seems likely that marked maternal toxicity contributed to the observed alterations in fetal growth retardation and skeletal development.

Pharmacokinetic parameters play a central role in establishing the amount and the form of the fungicide that reaches the embryonic compartment through placental transfer (Committee on Developmental Toxicology, 2000). It is known that animal species can differ significantly in the rates at which they absorb, distribute, and eliminate the pesticides and that these differences represent a major obstacle in the extrapolation of teratogenic data from laboratory animals to humans (Brent, 2004). Flusilazole is lipophilic compound, which results in its low free concentration in blood (Anderson et al., 1986; Cheng, 1986; Trösken and Main, 2005). It is one of a group of compounds known as sterol biosynthesis inhibitors. Flusilazole acts against ergosterol biosynthesis at the step lanosterol demethylase. It binds to a haem preventing demethylation and can also be known as a demethylation inhibitor. The result of binding is an increase in sterol precursors, with a consequent reduction in membrane integrity and altered structure. Cell multiplication is slowed severely and the altered cells are more susceptible to phagocytosis. Flusilazole also partially blocks cholesterol biosynthesis

(Yoshida, 1988; Zarn et al., 2003; Waterman and Lepe-shera, 2005; Trösken and Main, 2005). These effects may be associated with maternal toxicity that may lead to fetal growth retardation and skeletal malformations that are found in the present study at middle and high dose levels.

CONCLUSIONS

The present data show that flusilazole can produce teratogenic effects in mice at 20 and 40 mg/kg/day, which were maternally toxic doses. No evidence of maternal and developmental toxicity was observed at 10 mg/kg/day. The 20 mg/kg/day dose is about 20,000 times the acceptable daily intake for humans (ADI = 0.001 mg/kg/day) (Trösken and Main, 2005).

REFERENCES

- Aliverti V, Bonanomi L, Giavini E, Leone VG, Mariani L. 1979. The extent of fetal ossification as an index of delayed development in teratogenic studies on the rat. *Teratology* 20:237–242.
- Anderson JJ, Stadius MA, Schlueter DD. 1986. Metabolism of 14 C-DPX-H6573 in rats. Wilmington, DE: DuPont de Nemours and Co., Inc.
- Ariyuki F, Higaki K, Yasuda M. 1980. A study of fetal growth retardation in teratological tests: an examination of the relationship between body weight and ossification of coccygeal vertebrae in mouse and rat fetuses. *Teratology* 22:43–49.
- Ariyuki F, Ishihara H, Higaki K, Yasuda M. 1982. A study of fetal growth retardation in teratological tests: An examination of the relationship between body weight and ossification of the skeleton in rat fetuses. *Teratology* 26:263–267.
- Brent RL. 2004. Utilization of animal studies to determine the effects and human risks of environmental toxicants (drugs, chemicals, and physical agents). *Pediatrics* 113:984–995.
- Buchenaer H. 1987. Mechanism of action of triazole fungicides and related compounds. In: Lyr H, editor. New York: Wiley. p 205–231.
- Chahoud L, Paumgarten FJR. 2005. Relationships between fetal body weight of Wistar rats at term and the extent of skeletal ossification. *Braz J Med Biol Res* 38:565–575.
- Cheng T. 1986. Rat metabolism, study of [triazole-3- 14 C] DPX-H6573. Wilmington, DE: Haskell Laboratory for Toxicology and Industrial Medicine.
- Committee on Developmental Toxicology. 2000. Board on environmental studies and toxicology. Washington, DC: National Academy Press. p 58–84.
- Food and Agriculture Organization. 1981. FAO plant and protection papers. 42. Pesticide residues in food—Evaluations. p 521–523.
- Food and Agriculture Organization. 1983. FAP plant and protection papers. 61. Pesticide residues in food—Evaluations. p 447–450.
- Food and Agriculture Organization. 1989. Plant and protection papers. 100/2 Pesticide residues in food —Evaluations. Part II: Toxicology. p 117–139.
- Hayes WJ. 1994. Pesticides studies in man. Baltimore: Williams and Wilkins.
- Hollingsworth J. 1995. Fungicide studies offer little comfort. A Medical Mystery. Tampa, FL: Tampa Tribune Press. p 4.
- Lamontia CL, Staples RE, Alvarez L. 1984. Embryo-fetal toxicity and teratogenicity study of INH-6573-39 by gavage in the rat. Wilmington, DE: Haskell Laboratory for Toxicology and Industrial Medicine.
- Lehmann EL. 1975. Nonparametrics: statistical methods based on ranks. San Francisco: Holden-Day.
- Machemer L. 1976. Evaluation of embryotoxic and teratogenic effects on rats following oral administration. Wuppertal-Elberfeld, Germany: Bayer AG.
- Machera K. 1995. Developmental toxicity of cyproconazole, an inhibitor of fungal ergosterol biosynthesis, in the rat. *Bull Environ Contam Toxicol* 54:363–369.
- Menegola E, Broccia ML, Di Renzo F, Giavini E. 2001. Antifungal triazoles induce malformations in vitro. *Reprod Toxicol* 15:421–427.
- Menegola E, Broccia ML, Di Renzo F, Massa V, Giavini E. 2005. Study on the common teratogenic pathway elicited by the fungicides triazole-derivatives. *Toxicol In Vitro* 19:737–748.
- Norusis M. 1994. Statistical package for social sciences, Version 6 SPSS. New York: McGraw-Hill.
- Peters PWJ. 1965. Double staining of fetal skeletons for cartilage and bone. In: Wilson JG, Warkany J, editors. Chicago: University of Chicago Press. 153p.

- Salewski E. 1964. Staining method for a macroscopic test for implantation points in the uterus of the rat. *Naunyn Schmiedebergs Arch Exp Pathol Pharmacol* 247:367.
- Simes RJ. 1986. An improved Bonferroni procedure for multiple tests of significance. *Biometrika* 73:751–754.
- Solomon HM, Alvarez L, Staples RE, Hamill JC. 1984. Developmental toxicity study in rabbits given INH-6573 by gavage on days 7–19 of gestation. Wilmington, DE: Haskell Laboratory for Toxicology and Industrial Medicine.
- Trösken ER, Main FA. 2005. Toxicological evaluation of azole fungicides in agriculture and food chemistry. Dissertation zur Erlangung des naturwissenschaftlichen Doktorgrades der Bayerischen Julius-Maximilians-Universität Würzburg.
- Vanden Bossche H, Lauvers W, Willemsens G, Marichal P, Cools W. 1984. The molecular basis for the antifungal activities of N-substituted azole derivatives. In: Trinci A, Ryles J, editors. *British Mycological Society Press*. p 321–341.
- Vergieva T. 1990. Triazoles teratogenicity in rats. *Teratology* 42:27A–28A.
- Walpole RE, Meyers RH. 1978. *Probability and statistics for engineers and scientists*. New York: McMillan.
- Waterman MR, Lepesheva GI. 2005. Sterol 14 alpha-demethylase, an abundant and essential mixed-function oxidase. *Biochem Biophys Res Commun* 338:418–422.
- Wilson JG, Warkany J. 1965. *Teratology: principles and techniques*. Chicago: University of Chicago Press.
- Winer BJ. 1971. *Statistical principles in experimental design*. New York: McGraw-Hill.
- Wylie CN, Redgate D, Warheit DB, Chromey NC. 1985. Approximate lethal dose (ALD) of INH-6573-66 in male and female mice. Wilmington, DE: Haskell Laboratory for Toxicology and Industrial Medicine.
- Yoshida Y. 1988. Cytochrome P450 of fungi: primary target for azole antifungal agents. *Curr Top Med Mycol* 2:388–418.
- Zarn JA, Bruschweiler BJ, Schlatter JR. 2003. Azole fungicides affect mammalian steroidogenesis by inhibiting sterol 14 alpha-demethylase and aromatase. *Environ Health Perspect* 111:255–262.